Which Error Can the District Afford?

AP Statistics · Topic 6.7 (CED 3.8) · B: 2026-12-08 · E: 2027-01-15 · TEACHER KEY

CED TETHER

- **Skill 1.B** — Identify key and relevant information to answer a question or solve a problem.
- **Skill 3.A** — Determine relative frequencies, proportions, or probabilities using simulation or calculations.
- **Skill 4.A** — Make an appropriate claim or draw an appropriate conclusion.
- **Skill 4.B** — Interpret statistical calculations and findings to assign meaning or assess a claim.
- **EU UNC-5** — Probabilities of Type I and Type II errors influence inference.
- **LO UNC-5.A** — Identify Type I and Type II errors. [Skill 1.B]
- **EK UNC-5.A.1** — A Type I error occurs when the null hypothesis is true and is rejected (false positive).
- **EK UNC-5.A.2** — A Type II error occurs when the null hypothesis is false and is not rejected (false negative).
- Table of Errors:
- H_0 true
- H_a true
- Reject H_0
- Type I Error
- Correct Decision
- Fail to Reject H_0
- Correct Decision
- Type II Error
- **LO UNC-5.B** — Calculate the probability of a Type I and Type II errors. [Skill 3.A]
- **EK UNC-5.B.1** — The significance level, α, is the probability of making a Type I error, if the null hypothesis is true.
- **EK UNC-5.B.2** — The power of a test is the probability that a test will correctly reject a false null hypothesis.
- **EK UNC-5.B.3** — The probability of making a Type II error = 1 - power.
- **LO UNC-5.C** — Identify factors that affect the probability of errors in significance testing. [Skill 4.A]
- **EK UNC-5.C.1** — The probability of a Type II error decreases when any of the following occurs, provided the others do not change: i. Sample size(s) increases.
- ii. Significance level (α) of a test increases.
- iii. Standard error decreases.
- iv. True parameter value is farther from the null.
- **LO UNC-5.D** — Interpret Type I and Type II errors. [Skill 4.B]
- **EK UNC-5.D.1** — Whether a Type I or a Type II error is more consequential depends upon the situation.
- **EK UNC-5.D.2** — Since the significance level, α, is the probability of a Type I error, the consequences of a Type I error influence decisions about a significance level.

Essential question: How should the consequences of false positives and false negatives shape a testing plan?

DOK-3 skill: 4.B · **FRQ pattern:** `alpha-vs-power-which-error-tradeoff-fits-consequences`

DOK rationale: Part (c) translates operating probabilities into contextual consequences, weighs two stakeholders' competing priorities, critiques two absolute claims, and proposes a design change that improves detection without erasing the tradeoff.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in)DOK: 1 → 3
Minutes: 45

Teacher does

- Board slide up at the bell. Tally A, B, and need-more-information votes.
- Begin the video at minute 5; return at minute 33 to translate each probability into a decision and true state.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Choose a plan and name the prioritized error.
- Complete the follow-along, finish (a)–(c), revise, and turn in the sheet.

Questions to ask

- If $p = 0.10$, what decision creates the dangerous miss?
- What does the complement of 0.748 represent?
- What does Plan B trade per 1,000 tests?

Adult role
Solo teacher. Circulate 5 pairs. Require both consequences and both sides of the tradeoff.

[WARN] WATCH FOR

- Using $1 - \alpha$ as the probability of a Type II error.
- Claiming that lowering alpha lowers both error probabilities when n is fixed.
- Choosing solely from the larger power without acknowledging its false-positive cost.

SCORING PART (c) — E / P / I

Expected elements

- **computes-both-error-risks** (required) — Uses $\beta = 1 -$ power and Type I risk α
- **recommends-conditionally** (required) — Makes a stakeholder-linked, conditional recommendation
- **acknowledges-tradeoff** (required) — Contrasts 129 fewer misses with 40 more false replacements
- **critiques-both-claims** (required) — Explains why neither lower nor higher α is cost-free
- **proposes-sample-size-change** (required) — Raises n to improve power at fixed α

E	Quantifies both errors, recommends conditionally, states the tradeoff and missing costs, critiques both claims, and raises n.
P	Defends a plan but omits a consequence, critique, comparison, missing input, or design change.
I	Uses $1 - \alpha$ for β, calls either plan cost-free, or names a universal winner.

[STAR] TODAY'S PROBLEM — annotated

A district tests $H_0 : p = 0.05$ versus $H_a : p > 0.05$, where p is the late-alert proportion. Both plans use $n = 200$ and have the table's operating probabilities. Safety prioritizes avoiding retention when $p = 0.10$; Budget prioritizes avoiding replacement when $p = 0.05$. No common cost scale exists.

Dev: "Use A. Smaller α lowers both error risks."

Lin: "Use B. Its higher power has no cost when safety matters."

Test plans ($n = 200$)

Plan	α	Power at $p = 0.10$
A	0.010	0.748
B	0.050	0.877

FIRST TAKE (5 min)

Which plan would you try first? Name the office you prioritized and one competing cost.
Key: Not graded. Either plan needs a conditional defense; neither is cost-free.

Rules callout "ERROR PROBABILITIES TRADE OFF (keep on the page)" is printed on the student sheet.

DOK 1 **(a)** Describe a Type I error and a Type II error in this context. Identify what $P(\text{reject } H_0 \mid p = 0.10)$ represents.

Key: A Type I error rejects H_0 and concludes that more than 5 percent of alerts are late, causing the district to replace the system, when the true late rate is 0.05. A Type II error fails to reject H_0 and retains the system when its true late rate exceeds 0.05; at the specified alternative, the true rate is 0.10. The conditional rejection probability at $p = 0.10$ is the power against $p = 0.10$.

DOK 2 **(b)** For each plan, calculate β. Then give the expected number of Type I errors per 1,000 repeated tests when $p = 0.05$ and Type II errors per 1,000 repeated tests when $p = 0.10$.

Key: Plan A has $\beta = 1 - 0.748 = 0.252$; per 1,000 repetitions it produces about $1{,}000(0.010) = 10$ Type I errors when $p = 0.05$ and $1{,}000(0.252) = 252$ Type II errors when $p = 0.10$. Plan B has $\beta = 1 - 0.877 = 0.123$; it produces about $1{,}000(0.050) = 50$ Type I errors and $1{,}000(0.123) = 123$ Type II errors in the corresponding states.

DOK 3 **(c)** Adjudicate both claims. Explain why no plan is universally best; recommend one for a stated office using both error probabilities and their per-1,000 consequences. Name information needed for a district-wide choice and a change that improves power at fixed α.

Key: No plan is universally best without consequence weights. For Safety, B is defensible: power is 0.877 versus 0.748, so Type II risk is 0.123 versus 0.252—123 versus 252 misses per 1,000 at $p = 0.10$. For Budget, A is defensible: Type I risk is 0.010 versus 0.050—10 versus 50 false replacements per 1,000 at $p = 0.05$. Thus B trades 129 fewer misses for 40 more false replacements. Dev ignores that lower α raises Type II risk at fixed n; Lin hides B's false-positive cost. The district needs relative costs and likely true states. Increasing n can raise power and lower β while holding α fixed.

EXIT

One thing the video changed about my first take: ______________________________